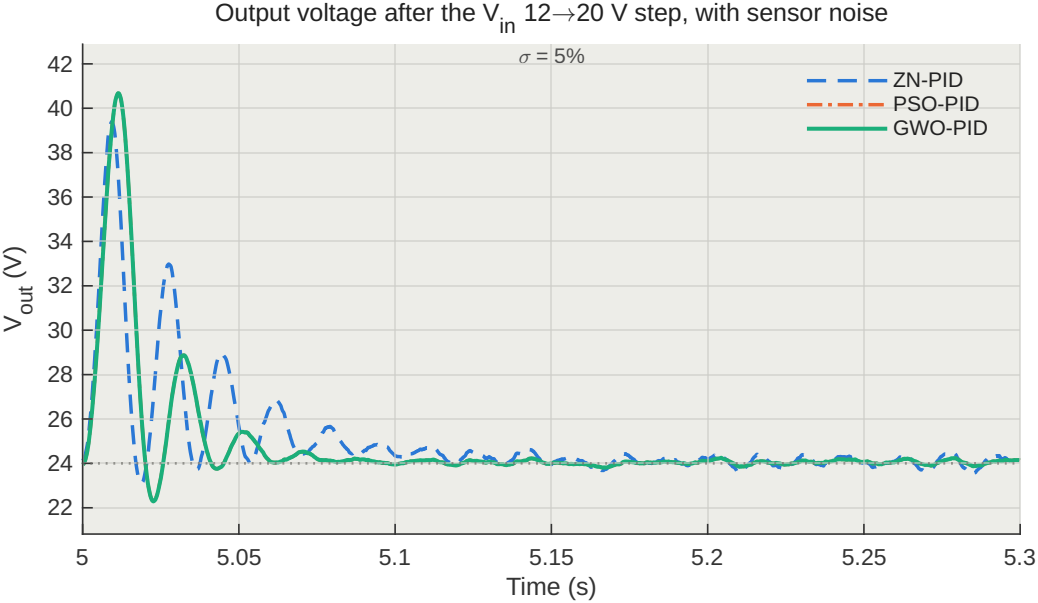

Output voltage after the V_{in} 12→20 V step, with sensor noise
σ = 5%
ZN-PID
PSO-PID
GWO-PID
V_{out} (V)
Time (s)
42
40
38
36
34
32
30
28
26
24
22
5
5.05
5.1
5.15
5.2
5.25
5.3